

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I C. Pavani/(192524260)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Pavani

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Question 1:

Q1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce

scenario.

Cloud Auto-Scaling and Load Balancing

Introduction

During seasonal sales such as festival offers or holiday discounts, websites receive a large number of users. If the servers cannot handle the increased traffic, the website may become slow or unavailable. Cloud computing solves this problem by using auto-scaling, load balancing, and monitoring tools to maintain high performance and availability.

1. Cloud Auto-Scaling

Auto-scaling automatically increases or decreases the number of cloud servers based on the current workload.

Working

During high traffic, new virtual servers are created automatically.

When traffic decreases, extra servers are removed.

Resources are used only when required.

Advantages

Handles sudden traffic spikes.

Prevents server overload.

Reduces operational cost.

Ensures continuous service availability.

2. Load Balancing

Load balancing distributes incoming user requests across multiple servers.

Working

Receives all user requests.

Checks which server has the lowest workload.

Sends each request to the most suitable server.

Benefits

Prevents one server from becoming overloaded.

Improves response time.

Increases system reliability.

Supports high availability.

3. How Auto-Scaling and Load Balancing Work Together

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Customers visit the website.

Load balancer distributes requests equally among servers.

Monitoring tools continuously observe CPU, memory, and network usage.

If server utilization becomes high, auto-scaling launches additional servers.

When demand reduces, unnecessary servers are removed automatically.

This combination ensures that applications remain fast and available even during peak traffic.

4. Reducing Downtime

Downtime occurs when servers fail to handle user requests.

Cloud services reduce downtime by:

Automatically adding more servers during heavy traffic.

Redirecting requests if one server fails.

Replacing failed servers quickly.

Maintaining continuous service without manual intervention.

As a result, users can access the website without interruptions.

5. Improving User Experience

Using cloud auto-scaling and load balancing provides:

Fast page loading.

Quick checkout process.

Smooth browsing experience.
Reduced waiting time.
Better customer satisfaction.
Increased customer trust.
Happy customers are more likely to complete purchases and return in the future.

6. Role of Monitoring Tools

Monitoring tools continuously collect information about system performance.

They monitor:

CPU utilization

Memory usage

Disk usage

Network traffic

Response time

Server health

Application performance

Functions

Detect performance issues.

Generate alerts for administrators.

Trigger auto-scaling policies.

Identify server failures.

Help improve overall system efficiency.

Monitoring ensures that cloud resources are managed effectively.

7. Real-World E-Commerce Scenario

Consider a large online shopping company conducting a festival sale.

Thousands of customers visit the website simultaneously.

The load balancer distributes customer requests across multiple servers.

Monitoring tools detect high CPU usage.

Auto-scaling automatically launches additional cloud instances.

Customers continue shopping without delays.

After the sale ends, extra servers are removed automatically to reduce costs.

This approach keeps the website available, improves customer experience, and prevents revenue loss.

Advantages

Automatic resource management

High availability

Better performance

Reduced downtime

Cost-efficient pay-as-you-use model

Improved scalability

Reliable customer experience

Conclusion

Cloud auto-scaling, load balancing, and monitoring tools work together to manage unpredictable traffic efficiently. Auto-scaling adjusts resources according to demand, load balancing distributes requests evenly, and monitoring tools continuously track system health. This combination minimizes downtime, improves application performance, reduces costs, and provides a smooth user experience during seasonal sales and other high-demand events.

Question 2:

Q2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Cloud Storage Solutions for Secure Backup Data

Introduction

Organizations generate large amounts of important data that must be backed up to prevent loss due to hardware failure, cyberattacks, or natural disasters. Cloud storage provides a secure, reliable, and cost-effective solution for storing backup data. It offers high durability, redundancy, encryption, access control, and disaster recovery capabilities.

1. Cloud Storage Solutions

Cloud storage allows organizations to store data on remote servers managed by cloud service providers. The data is accessible through the internet and can be expanded as storage needs grow.

Features

- Scalable storage capacity
- High availability
- Automatic backup
- Easy data access
- Pay-as-you-use pricing

2. Durability of Cloud Storage

Durability means data remains safe and is not lost even if hardware fails.

Cloud providers ensure durability by:

- Storing multiple copies of data.
- Using reliable storage devices.
- Automatically detecting and repairing damaged data.
- Regularly checking data integrity.

Benefits

- Prevents permanent data loss.
- Ensures long-term data protection.
- Provides reliable backup storage.

3. Data Redundancy

Data redundancy means storing multiple copies of the same data in different locations.

How it works

- Data is copied to multiple servers.
- Copies are stored in different data centers or regions.
- If one server fails, another copy is immediately available.

Advantages

- High availability
- Fault tolerance
- Faster recovery
- Continuous service during failures

4. Encryption for Data Protection

Encryption converts readable data into an unreadable format so that unauthorized users cannot access it.

Types

- Data at Rest: Encrypts stored backup files.
- Data in Transit: Encrypts data while it is transferred over the network.

Benefits

- Protects confidential information.
- Prevents unauthorized access.
- Improves data security.
- Supports regulatory compliance.

5. Access Control

Access control ensures that only authorized users can access backup data.

Methods

- User authentication
- Strong passwords
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Permission management

Benefits

- Prevents unauthorized access.
- Protects sensitive information.
- Improves overall security.
- Reduces insider threats.

6. Cost Advantages Compared to Traditional Storage

Cloud storage is more economical than traditional on-premises storage.

Traditional Storage

- High hardware costs
- Maintenance expenses
- Limited scalability
- Requires physical storage space
- Higher electricity and cooling costs

Cloud Storage

- Pay only for the storage used.
- No hardware purchase required.
- Easy to increase or decrease storage.
- Lower maintenance costs.
- Automatic software updates managed by the provider.

Advantages

- Lower initial investment
- Reduced operational costs
- Flexible pricing
- Better resource utilization

7. Disaster Recovery Use Case

A company stores all customer records and business files in cloud backup storage.

- A flood damages the company's local servers.
- All local data becomes unavailable.
- The organization accesses the latest backup stored in the cloud.
- Data is restored to new servers.
- Business operations continue with minimal downtime.

This demonstrates how cloud storage supports effective disaster recovery and business continuity.

Advantages of Cloud Backup Storage

- High durability
- Data redundancy
- Secure encryption
- Strong access control
- Automatic backups
- Easy scalability
- Cost-effective storage
- Fast disaster recovery
- High availability
- Reliable data protection

Conclusion

Cloud storage provides a secure and efficient solution for storing large volumes of backup data. Durability and redundancy protect data from hardware failures, while encryption and access control safeguard sensitive information. Compared to traditional storage systems, cloud storage reduces costs, improves scalability, and simplifies management. It also enables fast disaster recovery, ensuring business continuity during unexpected failures.

Question 3

Q3. A mobile app company plans to deploy its application globally. Explain how Content Delivery Networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Content Delivery Networks (CDNs) and Global Application Deployment

Introduction

A mobile application with users across different countries must provide fast and reliable access. If all users connect to a single server, they may experience delays. Cloud computing solves this problem by using Content Delivery Networks (CDNs), edge servers, and global cloud infrastructure to improve performance and availability.

1. Content Delivery Network (CDN)

A Content Delivery Network (CDN) is a group of servers located in different geographical regions that store copies of application content and deliver it to users from the nearest server.

How a CDN Works

1. The application content is stored on multiple CDN servers.
2. When a user requests data, the request is sent to the nearest CDN server.
3. The server delivers the content quickly without contacting the main server every time.

Benefits

- Faster content delivery.
- Reduced network congestion.
- Lower server workload.
- Better application performance.
- High availability.

2. How CDNs Reduce Latency

Latency is the time taken for data to travel between the user and the server.

CDNs reduce latency by:

- Delivering content from the nearest server.
- Reducing the distance data travels.
- Decreasing response time.
- Reducing network delays.

Advantages

- Faster page loading.
- Smooth application performance.
- Improved user satisfaction.

3. Edge Servers and User Experience

Edge servers are servers located close to users in different regions.

Role of Edge Servers

- Store frequently accessed content.
- Deliver data quickly to nearby users.
- Reduce the load on the central server.
- Improve response speed.
- Continue serving users even if one server experiences problems.

Benefits

- Faster downloads.
- Smooth video playback.
- Reduced buffering.
- Better gaming experience.
- Consistent performance worldwide.

4. Global Availability Through Cloud Providers

Cloud providers maintain data centers in many countries and regions.

They support global availability by:

- Deploying applications across multiple regions.
- Automatically routing users to the nearest server.
- Replicating application data across regions.
- Providing backup servers for fault tolerance.
- Ensuring high availability even during failures.

Advantages

- 24/7 service availability.
- Reliable application access.
- Faster disaster recovery.
- Better scalability.

5. Real-Time Streaming or Gaming Application

Consider a global online gaming application.

- Players connect from different countries.
- CDN delivers game updates and static content from nearby servers.
- Edge servers reduce communication delay between players and servers.
- Cloud infrastructure automatically handles increased traffic during tournaments.
- Players experience low latency, smooth gameplay, and minimal interruptions.

Similarly, in a video streaming application, users can watch videos with faster loading and less buffering because the content is delivered from the nearest edge server.

Advantages of CDN and Edge Servers

- Reduced latency.
- Faster content delivery.
- Improved application performance.
- Better user experience.
- High availability.
- Reduced bandwidth usage.
- Lower load on the main server.
- Easy global scalability.
- Reliable service during peak traffic.

Conclusion

Content Delivery Networks (CDNs), edge servers, and global cloud infrastructure help mobile applications deliver fast and reliable services worldwide. CDNs reduce latency by serving content from nearby servers, while edge servers improve response time and user experience. Cloud providers ensure global availability through multiple data centers, automatic scaling, and fault tolerance. These technologies are essential for real-time streaming and gaming applications, providing high performance, reliability, and customer satisfaction.

Question 4:

Q4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Introduction

Financial institutions handle sensitive customer information such as bank account details, credit card numbers, and transaction records. Therefore, they require strong security, regulatory compliance, and continuous monitoring when using cloud services. Cloud providers offer security features that help organizations protect data and meet legal requirements.

1. Regulatory Compliance

Regulatory compliance means following government laws and industry standards for storing and processing data.

Cloud providers support compliance by:

- Providing secure cloud infrastructure.
- Following international security standards.
- Performing regular security audits.
- Maintaining compliance certifications.
- Offering compliance reports to customers.

Benefits

- Protects customer information.
- Meets legal requirements.
- Reduces security risks.
- Builds customer trust.

2. Cloud Auditing

Auditing is the process of reviewing cloud activities and security controls.

Cloud auditing helps to:

- Track user activities.
- Detect unauthorized access.
- Verify compliance with regulations.
- Identify security weaknesses.
- Generate audit reports.

Advantages

- Improves transparency.
- Supports investigations.
- Enhances security management.
- Ensures accountability.

3. Role of Logging

Logging records every activity performed in the cloud.

Examples include:

- User login and logout.
- File access.
- Data modifications.
- Transactions.
- System errors.

Benefits

- Detects suspicious activities.
- Helps investigate security incidents.
- Maintains complete activity records.
- Supports auditing.

4. Role of Monitoring

Monitoring continuously checks the health and security of cloud resources.

It monitors:

- CPU usage.
- Memory utilization.
- Network traffic.
- Server health.
- Security threats.
- System performance.

Benefits

- Detects attacks quickly.
- Sends real-time alerts.
- Prevents service failures.
- Improves system reliability.

5. Role of Encryption

Encryption converts readable data into an unreadable format that can only be accessed using the correct decryption key.

Types of Encryption

- Data at Rest: Protects stored financial data.
- Data in Transit: Protects data while it is transmitted over networks.

Benefits

- Prevents unauthorized access.
- Protects confidential financial information.
- Ensures secure communication.
- Supports regulatory compliance.

6. Data Sovereignty

Data sovereignty means that an organization's data must remain stored and processed according to the laws of a specific country.

How to Ensure Data Sovereignty

- Store data in cloud data centers located within the required country.
- Select the appropriate cloud region.
- Follow local data protection regulations.
- Regularly verify where data is stored.
- Use cloud provider tools to control data location.

Advantages

- Meets legal requirements.
- Protects customer privacy.
- Reduces compliance risks.
- Improves trust.

7. Example: Sensitive Financial Transactions

Consider a bank that processes online fund transfers.

- Customer transaction data is encrypted before storage and during transmission.
- Every login and transaction is recorded in logs.
- Monitoring tools detect unusual activities, such as repeated failed login attempts.
- Audit reports help verify compliance with banking regulations.
- Customer data is stored in a cloud region within the country to satisfy data sovereignty requirements.

This ensures secure, reliable, and compliant financial transactions.

Advantages of Cloud Security

- Strong data protection.
- Secure encryption.
- Continuous monitoring.
- Detailed logging.
- Regulatory compliance.
- Secure auditing.

- High availability.
- Customer trust.
- Reduced security risks.
- Better business continuity.

Conclusion

Cloud providers help financial institutions achieve high levels of security and regulatory compliance through auditing, logging, monitoring, and encryption. These features protect sensitive financial information, detect threats, and maintain secure operations. By storing data in approved cloud regions, organizations can also meet data sovereignty requirements. Together, these measures ensure safe, reliable, and legally compliant cloud services for financial transactions.

Sources

Question 5:

Q5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Introduction

Modern software development requires applications to be deployed quickly, efficiently, and reliably. Containerization packages an application along with its dependencies into a lightweight container, making deployment easier across different environments. Kubernetes helps manage these containers by automating deployment, scaling, and monitoring.

1. What is Containerization?

Containerization is a technology that packages an application and all its required libraries, files, and dependencies into a single container.

Features

- Lightweight
- Fast startup
- Portable
- Efficient resource usage
- Consistent application environment

2. Containers vs Virtual Machines

Containers	Virtual Machines (VMs)
Share the host operating system	Each VM has its own operating system
Lightweight	Heavyweight
Start in seconds	Take several minutes to start
Use less CPU and memory	Consume more CPU and memory
High performance	Lower performance compared to containers
More containers can run on one server	Fewer VMs can run on one server

3. Performance and Resource Usage

Containers

- Fast startup time.
- Low memory usage.
- Better CPU utilization.
- Higher application performance.
- Suitable for cloud-native applications.

Virtual Machines

- Require a complete operating system.
- Higher memory and storage usage.
- Slower startup.
- More hardware resources needed.

Result: Containers provide better performance and more efficient resource utilization than virtual machines.

4. Kubernetes Orchestration

Kubernetes is an orchestration platform that manages multiple containers automatically.

Functions of Kubernetes

- Deploys applications automatically.
- Scales containers based on user demand.
- Restarts failed containers.
- Balances network traffic.
- Performs rolling updates without downtime.
- Monitors container health.

5. How Kubernetes Simplifies Deployment and Scaling

Kubernetes simplifies application management by:

- Automatically deploying containers.
- Increasing or decreasing the number of containers based on traffic.
- Replacing failed containers automatically.
- Distributing workloads across servers.
- Supporting continuous updates with minimal downtime.

Benefits

- High availability.
- Easy management.
- Automatic recovery.
- Improved reliability.
- Better scalability.

6. Portability Across Different Cloud Environments

Containers can run on different cloud platforms without changing the application code.

Benefits

- Deploy once and run anywhere.
- Easy migration between cloud providers.
- Consistent performance.
- Reduced vendor lock-in.
- Simplified testing and deployment.

This allows organizations to move applications between private, public, or hybrid clouds with minimal effort.

7. Example of a Microservices-Based Application

An online shopping (e-commerce) application can be built using microservices.

Each service runs in its own container:

- User Service
- Product Service
- Order Service
- Payment Service
- Inventory Service
- Notification Service

Kubernetes manages these containers by:

- Scaling the Order Service during festival sales.
- Restarting failed services automatically.
- Balancing user requests across multiple containers.
- Ensuring continuous availability.

This architecture improves performance, reliability, and scalability.

Advantages of Containerization

- Lightweight deployment.
- Faster startup.
- Efficient resource utilization.
- Easy application portability.
- Improved scalability.
- High performance.
- Simplified application management.
- Better fault isolation.
- Faster software updates.
- Reduced infrastructure costs.

Conclusion

Containerization provides a fast, lightweight, and efficient way to deploy applications, while virtual machines require more resources and startup time. Kubernetes simplifies container deployment, scaling, monitoring, and recovery through automation. Containers also provide excellent portability across different cloud environments, allowing organizations to deploy applications consistently on any cloud platform. A microservices-based architecture, such as an e-commerce application, demonstrates how containers and Kubernetes improve scalability, reliability, and overall application performance.